Projectile Motion

1. A ball is launched straight upward with an initial velocity of 20 m/s. What is the time to reach maximum height?

Given:

$V_i = 20\, m/s$
$g = 10\, m/s^2$

At maximum height, the final velocity is zero.

$$t = \frac{V_i}{g}$$

$$t = \frac{20}{10} = 2\, s$$

Answer: B. 2 s

2. A ball is launched straight upward at 30 m/s. What is the maximum height reached?

Given:

$V_i = 30\, m/s$
$g = 10\, m/s^2$

Use the maximum-height formula:

$$h = \frac{V_i^2}{2g}$$

$$h = \frac{30^2}{2(10)}$$

$$h = \frac{900}{20} = 45\, m$$

Answer: C. 45 m

3. A projectile is launched with a vertical component of velocity of 20 m/s and lands at the same height. What is its total time of flight?

Given:

$V_y = 20\, m/s$
$g = 10\, m/s^2$

First, find the time to reach maximum height:

$$t_{up} = \frac{V_y}{g}$$

$$t_{up} = \frac{20}{10} = 2\, s$$

Since the projectile lands at the same height:

$$T = 2t_{up}$$

$$T = 2(2) = 4\, s$$

Answer: C. 4 s

4. A projectile moves horizontally at 15 m/s for 4 seconds. What is its range?

Given:

$V_x = 15\, m/s$
$t = 4\, s$

Use:

$$R = V_x t$$

$$R = 15(4) = 60\, m$$

Answer: C. 60 m

5. A ball is launched straight upward at 40 m/s. What is its time to reach maximum height?

Given:

$V_i = 40\,m/s$
$g = 10\,m/s^2$

$$t = \frac{V_i}{g}$$

$$t = \frac{40}{10} = 4\,s$$

Answer: C. 4 s

6. A projectile has a horizontal velocity of 12 m/s and stays in the air for 5 seconds. What is its range?

Given:

$V_x = 12\,m/s$
$t = 5\,s$

$$R = V_x t$$

$$R = 12(5) = 60\,m$$

Answer: C. 60 m

7. A ball is thrown upward with an initial vertical velocity of 10 m/s. What is its maximum height?

Given:

$V_i = 10\,m/s$
$g = 10\,m/s^2$

$$h = \frac{V_i^2}{2g}$$

$$h = \frac{10^2}{2(10)}$$

$$h = \frac{100}{20} = 5\,m$$

Answer: A. 5 m

8. A projectile takes 6 seconds to reach the ground after being launched and lands at the same height. What is the time to reach maximum height?

Given:

$T = 6\,s$

Because the projectile lands at the same height, the upward and downward times are equal.

$$t_{up} = \frac{T}{2}$$

$$t_{up} = \frac{6}{2} = 3\,s$$

Answer: B. 3 s

9. A projectile has a horizontal velocity of 20 m/s and a time of flight of 3 seconds. What is its range?

Given:

$V_x = 20\,m/s$

$T = 3\,s$

$$R = V_x T$$

$$R = 20(3) = 60\,m$$

Answer: C. 60 m

10. A ball is launched straight upward at 50 m/s. What is the maximum height?

Given:

$V_i = 50\,m/s$

$g = 10\,m/s^2$

$$h = \frac{V_i^2}{2g}$$

$$h = \frac{50^2}{2(10)}$$

$$h = \frac{2500}{20} = 125\,m$$

Answer: D. 125 m

11. A projectile is launched with a vertical velocity of 40 m/s and lands at the same level. What is the time of flight?

Given:
$V_y = 40\,m/s$

Time to maximum height:

$$t_{up} = \frac{40}{10} = 4\,s$$

Total time of flight:

$$T = 2(4) = 8\,s$$

Answer: C. 8 s

12. A projectile moves horizontally at 25 m/s and stays in the air for 2 seconds. What is the range?

Given:
$V_x = 25\,m/s$
$T = 2\,s$

$$R = V_x T$$

$$R = 25(2) = 50\,m$$

Answer: B. 50 m

13. A ball is launched vertically upward at 60 m/s. What is the time to reach maximum height?

Given:
$V_i = 60\,m/s$

$$t = \frac{V_i}{g}$$

$$t = \frac{60}{10} = 6\,s$$

Answer: D. 6 s

14. A projectile is launched vertically upward at 60 m/s. What is its maximum height?

Given:

$V_i = 60\, m/s$

$$h = \frac{V_i^2}{2g}$$

$$h = \frac{60^2}{2(10)}$$

$$h = \frac{3600}{20} = 180\, m$$

Answer: C. 180 m

15. A projectile has a horizontal velocity of 18 m/s and stays in the air for 4 seconds. What is its range?

Given:

$V_x = 18\, m/s$

$T = 4\, s$

$$R = V_x T$$

$$R = 18(4) = 72\, m$$

Answer: C. 72 m

16. A projectile is launched with a vertical velocity of 30 m/s and lands at the same height. What is the time of flight?

Given:

$V_y = 30\, m/s$

Time to maximum height:

$$t_{up} = \frac{30}{10} = 3\, s$$

Total time:

$$T = 2(3) = 6\, s$$

Answer: C. 6 s

17. A ball is launched vertically upward with an initial velocity of 70 m/s. What is its maximum height?

Given:

$V_i = 70\,m/s$

$$h = \frac{V_i^2}{2g}$$

$$h = \frac{70^2}{2(10)}$$

$$h = \frac{4900}{20} = 245\,m$$

Answer: C. 245 m

18. A projectile travels horizontally at 30 m/s for 5 seconds. What is the range?

Given:

$V_x = 30\,m/s$
$T = 5\,s$

$$R = V_x T$$

$$R = 30(5) = 150\,m$$

Answer: C. 150 m

19. A projectile stays in the air for 10 seconds and lands at the same height where it was launched. What is the time to reach maximum height?

Given:

$T = 10\,s$

Since the projectile lands at the same height:

$$t_{up} = \frac{T}{2}$$

$$t_{up} = \frac{10}{2} = 5\,s$$

Answer: B. 5 s

20. A projectile has a horizontal velocity of 14 m/s and remains in the air for 6 seconds. What is its range?

Given:
$V_x = 14\, m/s$
$T = 6\, s$

$$R = V_x T$$

$$R = 14(6) = 84\, m$$

Answer: D. 84 m

Important Formulas

Time to reach maximum height:

$$\boxed{t_{up} = \frac{V_i}{g}}$$

Maximum height:

$$\boxed{h_{max} = \frac{V_i^2}{2g}}$$

Total time of flight when landing at the same height:

$$\boxed{T = 2t_{up}}$$

Horizontal range:

$$\boxed{R = V_x T}$$

Use:

$$\boxed{g = 10\, m/s^2}$$

Power

1. What is the formula for power?

A. $P = F \times d$
B. $P = W \div t$
C. $P = m \times v$
D. $P = E \times t$

Formula:

$$\boxed{P = \frac{W}{t}}$$

Power is the rate at which work is done.

Answer: B. $P = W \div t$

Answer: C. 30 V

2. The SI unit of power is:

A. Joule
B. Newton
C. Watt
D. Meter

The SI unit used to measure power is the **watt (W)**.

$$\boxed{1\,W = 1\,J/s}$$

Answer: C. Watt

3. One watt is equal to:

A. 1 joule per second

B. 1 newton per meter

C. 1 meter per second

D. 1 kilogram per second

A watt measures power. One watt means that **one joule of work is done every second.**

$$\boxed{1\,W = 1\,J/s}$$

Answer: A. 1 joule per second

4. If the same amount of work is done in less time, the power is:

A. Less

B. Greater

C. Zero

D. Unchanged

Use:

$$P = \frac{W}{t}$$

If the work stays the same but the time becomes smaller, the power becomes greater.

Answer: B. Greater

5. A machine does 100 J of work in 5 seconds. What is its power?

A. 20 W

B. 50 W

C. 100 W

D. 500 W

Given:

$$W = 100\,J, \quad t = 5\,s$$

Use:

$$P = \frac{W}{t}$$

$$P = \frac{100}{5} = 20\,W$$

Answer: A. 20 W

6. Which formula relates power, force, and velocity?

A. $P = Fv$

B. $P = F \div v$

C. $P = F + v$

D. $P = v \div F$

When force and velocity are in the same direction:

$$\boxed{P = Fv}$$

Answer: A. $P = Fv$

7. A force of 10 N moves an object at 3 m/s in the same direction. What is the power?

A. 3 W

B. 13 W

C. 30 W

D. 300 W

Given:

$$F = 10\,N, \quad v = 3\,m/s$$

Use:

$$P = Fv$$

$$P = 10(3) = 30\,W$$

Answer: C. 30 W

8. Power measures how fast:

A. An object moves

B. Work is done

C. Force is applied

D. Mass increases

Power tells us **how quickly work is done.**

$$\boxed{P = \frac{W}{t}}$$

Answer: B. Work is done

9. Which quantity is measured in joules?

A. Power

B. Force

C. Work and energy

D. Velocity

The **joule (J)** is the SI unit for both work and energy.

$$\boxed{\text{Work and Energy} = J}$$

Answer: C. Work and energy

10. What is the unit of force?

A. Watt

B. Joule

C. Newton

D. Meter per second

The SI unit of force is the **newton (N)**.

$$\boxed{\text{Force} = N}$$

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11. If force and velocity are in the same direction, power can be calculated by:

A. $P = F \times v$

B. $P = F + v$

C. $P = F - v$

D. $P = F \div v$

When force and velocity have the same direction:

$$\boxed{P = Fv}$$

Answer: A. $P = F \times v$

12. A person does 600 J of work in 20 seconds. What is the power?

A. 20 W

B. 30 W

C. 40 W

D. 50 W

Given:

$$W = 600\,J, \quad t = 20\,s$$

Use:

$$P = \frac{W}{t}$$

$$P = \frac{600}{20} = 30\,W$$

Answer: B. 30 W

13. Which statement is TRUE about power and work?

A. Power is the total amount of work.

B. Power is the rate at which work is done.

C. Work and power have the same unit.

D. Power is measured in newtons.

Power describes **how quickly work is performed.**

$$\boxed{P = \frac{W}{t}}$$

Answer: B. Power is the rate at which work is done.

14. If an object moves faster while the same force is applied, the power will:

A. Decrease

B. Stay the same

C. Increase

D. Become zero

Use:

$$P = Fv$$

If the force stays the same and velocity increases, power also increases.

Answer: C. Increase

15. A machine uses 200 W of power for 10 seconds. How much energy does it transfer?

A. 20 J

B. 200 J

C. 2,000 J

D. 20,000 J

Given:

$$P = 200\,W, \quad t = 10\,s$$

Use:

$$E = Pt$$

$$E = 200(10) = 2,000\,J$$

Answer: C. 2,000 J

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16. Which equation can be used to calculate energy using power and time?

A. $E = P \times t$

B. $E = P \div t$

C. $E = F \times v$

D. $E = F \div t$

Energy can be calculated from power and time using:

$$\boxed{E = Pt}$$

Answer: A. $E = P \times t$

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17. A motor exerts a force of 50 N and moves at 4 m/s. What is its power?

A. 54 W

B. 200 W

C. 250 W

D. 400 W

Given:

$$F = 50\,N, \quad v = 4\,m/s$$

Use:

$$P = Fv$$

$$P = 50(4) = 200\,W$$

Answer: B. 200 W

18. Two students do the same amount of work. Student A finishes faster than Student B. Who has greater power?

A. Student A

B. Student B

C. Both have the same power

D. Cannot be determined

Use:

$$P = \frac{W}{t}$$

Both students do the same work, but Student A takes less time. Therefore, Student A has greater power.

Answer: A. Student A

19. Power is directly related to:

A. Work done per unit time

B. Mass only

C. Distance only

D. Time only

Power is defined as the amount of work done per unit of time.

$$\boxed{P = \frac{W}{t}}$$

Answer: A. Work done per unit time

20. Which equation correctly shows the relationship between power, work, and time?

A. $P = W \times t$

B. $P = W \div t$

C. $P = t \div W$

D. $P = W + t$

The correct formula for power is:

$$\boxed{P = \frac{W}{t}}$$

Answer: B. $P = W \div t$

Important Formulas

Power from work and time:

$$\boxed{P = \frac{W}{t}}$$

Power from force and velocity:

$$\boxed{P = Fv}$$

Energy from power and time:

$$\boxed{E = Pt}$$

SI Units:

- Power = **watt (W)**
- Work = **joule (J)**
- Energy = **joule (J)**
- Force = **newton (N)**
- Time = **second (s)**
- Velocity = **meter per second (m/s)**

NEWTON'S LAWS OF MOTION

1.Newton's First Law of Motion is also known as the Law of:

A. Acceleration
B. Inertia
C. Action-Reaction
D. Gravitation

Newton's First Law states that an object tends to remain at rest or continue moving at constant velocity unless acted upon by an unbalanced force. This property is called inertia.

Answer: B. Inertia

2. A 10 kg object is pushed with a net force of 50 N. What is its acceleration?

A. 2 m/s^2
B. 5 m/s^2
C. 10 m/s^2
D. 500 m/s^2

Given:

$m = 10$ kg
$F = 50$ N

Formula:

$F = ma$

Solve for acceleration:

$a = F \div m$

$a = 50 \div 10$

$a = 5\ m/s^2$

Answer: B. 5 m/s^2

3. Which equation represents Newton's Second Law of Motion?

A. $F = mv$
B. $F = ma$
C. $W = mg$
D. $v = d/t$

Newton's Second Law relates force, mass, and acceleration:

$F = ma$

Answer: B. $F = ma$

4. A box remains at rest on a table. Which statement is correct?

A. No forces are acting on it.
B. The net force acting on it is zero.
C. Gravity is not acting on it.
D. The box has no mass.

The box is at rest because the forces acting on it are balanced. Therefore, the net force is zero.

Answer: B. The net force acting on it is zero.

5. A person pulls a rope attached to a box with a force of 100 N. If friction is 30 N, what is the net force on the box?

A. 30 N
B. 70 N
C. 100 N
D. 130 N

Given:

Applied force = 100 N
Friction = 30 N

Since friction acts opposite the pulling force:

Net force = 100 − 30

Net force = 70 N

Answer: B. 70 N

6. In an ideal pulley system, the main purpose of a pulley is to:

A. Increase the mass of an object
B. Change the direction or reduce the effort needed to lift a load
C. Remove gravity
D. Stop an object from moving

A pulley can change the direction of the applied force and, in some arrangements, reduce the effort needed to lift a load.

Answer: B. Change the direction or reduce the effort needed to lift a load

7. A 20 kg load is lifted upward with a constant velocity. What is the net force on the load?

A. 0 N
B. 20 N
C. 196 N
D. 392 N

Constant velocity means the acceleration is zero.

Using:

$F = ma$

$F = 20(0)$

$F = 0$ N

Answer: A. 0 N

8. A rope is used to pull two boxes in a straight line. The force transmitted through the rope is called:

A. Friction
B. Weight
C. Tension
D. Momentum

The pulling force transmitted through a rope or string is called tension.

Answer: C. Tension

9. A 5 kg object hangs from a rope and is at rest. Using $g = 9.8\ m/s^2$, what is the tension in the rope?

A. 5 N
B. 9.8 N
C. 49 N
D. 98 N

Given:

$m = 5$ kg
$g = 9.8\ m/s^2$

Weight:

$W = mg$

$W = 5(9.8)$

$W = 49$ N

Since the object is at rest, the tension balances the weight.

$T = 49$ N

Answer: C. 49 N

10. If the upward tension force on a hanging object is greater than its weight, the object will:

A. Accelerate upward
B. Accelerate downward
C. Remain at rest only
D. Lose its mass

If the upward force is greater than the downward weight, there is a net upward force.

Therefore, the object accelerates upward.

Answer: A. Accelerate upward

11. A mobile sculpture is balanced when:

A. All objects have the same mass.
B. The net torque on each balanced support is zero.
C. All objects are at the same height.
D. Gravity does not act on the sculpture.

A balanced mobile has zero net torque around its pivot.

Net torque = 0

Answer: B. The net torque on each balanced support is zero.

12. Torque is calculated by:

A. Force × perpendicular distance
B. Mass × acceleration
C. Mass × gravity
D. Distance ÷ time

The formula for torque is:

Torque = Force × perpendicular distance

$\tau = Fd$

Answer: A. Force × perpendicular distance

13. A 10 N weight is placed 2 m from the pivot of a mobile. What is the torque produced?

A. 5 N·m
B. 10 N·m
C. 20 N·m
D. 40 N·m

Given:

F = 10 N
d = 2 m

Formula:

$\tau = Fd$

$\tau = 10(2)$

$\tau = 20$ N·m

Answer: C. 20 N·m

14. A mobile sculpture has a 20 N weight located 1 m to the left of the pivot. Where should a 10 N weight be placed on the right to balance it?

A. 0.5 m
B. 1 m
C. 2 m
D. 4 m

For balance, the clockwise and counterclockwise torques must be equal.

Left torque:

$\tau = 20(1)$

$\tau = 20 \text{ N·m}$

For the 10 N weight:

$20 = 10d$

$d = 20 \div 10$

$d = 2 \text{ m}$

Answer: C. 2 m

15. Why can a lighter object balance a heavier object on a mobile sculpture?

A. It has more gravity.
B. It can be placed farther from the pivot.
C. It has no weight.
D. It removes friction.

Torque depends on both force and distance from the pivot.

$\tau = Fd$

A lighter object can produce the same torque as a heavier object if it is placed farther from the pivot.

Answer: B. It can be placed farther from the pivot.

16. A conveyor belt moves a 50 kg load with a net horizontal force of 100 N. What is the acceleration of the load?

A. 0.5 m/s^2
B. 2 m/s^2
C. 50 m/s^2
D. 150 m/s^2

Given:

$F = 100$ N
$m = 50$ kg

Formula:

$a = F \div m$

$a = 100 \div 50$

$a = 2\ m/s^2$

Answer: B. 2 m/s^2

17. Friction between a conveyor belt and a load helps the load to:

A. Fall through the belt
B. Move together with the belt
C. Lose its mass
D. Stop gravity

Friction between the belt and the load provides the force needed for the load to move along with the belt.

Answer: B. Move together with the belt

18. If a conveyor belt suddenly stops, a load may continue moving forward due to:

A. Inertia
B. Torque
C. Buoyancy
D. Magnetism

According to Newton’s First Law, an object in motion tends to remain in motion unless acted upon by an unbalanced force.

This property is called inertia.

Answer: A. Inertia

19. A load on a conveyor belt moves at constant velocity. What can be said about the net force?

A. It is increasing.
B. It is zero.
C. It is equal to the mass.
D. It is always downward.

Constant velocity means the acceleration is zero.

Using:

$F = ma$

$F = m(0)$

$F = 0\ N$

Answer: B. It is zero.

20. Which change would require a greater force to accelerate a load on a conveyor belt?

A. Decreasing its mass
B. Increasing its mass
C. Reducing the acceleration to zero
D. Removing all forces

According to Newton’s Second Law:

F = ma

If mass increases while the desired acceleration remains the same, a greater force is needed.

Answer: B. Increasing its mass

21. A stalled vehicle requires a force of 2,000 N to move. If four people push with equal force, how much force must each person exert, assuming no other forces?

A. 250 N
B. 500 N
C. 1,000 N
D. 2,000 N

Total force needed = 2,000 N

Number of people = 4

Force per person:

F = 2,000 ÷ 4

F = 500 N

Answer: B. 500 N

22. Two people push a stalled car in the same direction with forces of 400 N and 600 N. What is the total applied force?

A. 200 N
B. 500 N
C. 1,000 N
D. 2,400 N

Since both forces act in the same direction, add them:

$F_{total} = 400 + 600$

$F_{total} = 1{,}000$ N

Answer: C. 1,000 N

23. A car is pushed with 1,500 N forward, while friction and resistance total 500 N backward. What is the net force?

A. 500 N backward
B. 1,000 N forward
C. 1,500 N forward
D. 2,000 N forward

Opposing forces are subtracted:

$F_{net} = 1{,}500 - 500$

$F_{net} = 1{,}000$ N

The larger force is forward, so the net force is 1,000 N forward.

Answer: B. 1,000 N forward

24. Why is it harder to start moving a stalled vehicle than to keep it moving?

A. Static friction must first be overcome.
B. Gravity disappears while it moves.
C. The mass decreases after moving.
D. Newton’s laws stop applying.

Static friction resists the initial motion of an object. A sufficiently large force must overcome static friction before the vehicle begins moving.

Answer: A. Static friction must first be overcome.

25. According to Newton’s Third Law, when a person pushes a car, the car:

A. Does not push back.
B. Pushes the person with an equal and opposite force.
C. Has no reaction force.
D. Pushes with twice the force.

Newton’s Third Law states that for every action force, there is an equal and opposite reaction force.

Answer: B. Pushes the person with an equal and opposite force.

26. A car travels around a banked curve. The banking helps provide:

A. Centripetal force
B. Buoyant force

C. Magnetic force
D. Nuclear force

A banked road helps provide a component of force directed toward the center of the curve. This helps provide the centripetal force needed for circular motion.

Answer: A. Centripetal force

27. What may happen if a vehicle travels too fast for a particular banked curve?

A. It may slide upward on the bank.
B. It will lose all its weight.
C. Gravity will stop acting on it.
D. The road will become flat.

If the vehicle travels faster than the speed the banked curve is designed for, the vehicle may slide upward along the bank.

Answer: A. It may slide upward on the bank.

28. What is the main reason roads are banked on some curves?

A. To make the road longer
B. To help vehicles turn safely
C. To decrease the mass of vehicles
D. To eliminate all friction

Banking helps provide the force needed to keep vehicles moving safely around a curve.

Answer: B. To help vehicles turn safely

29. On a properly designed banked curve, the normal force from the road has a horizontal component that:

A. Provides centripetal force
B. Cancels gravity completely
C. Stops the vehicle permanently
D. Increases the vehicle’s mass

The horizontal component of the normal force points toward the center of the curve and provides centripetal force.

Answer: A. Provides centripetal force

30. A car travels around a curve of radius 50 m. If the safe speed increases, what happens to the required centripetal force, assuming the mass and radius remain constant?

A. It decreases.
B. It remains the same.
C. It increases.
D. It becomes zero.

Centripetal force is given by:

$F_c = mv^2 \div r$

If mass and radius remain constant, increasing the speed increases the centripetal force.

Answer: C. It increases.

IMPORTANT FORMULAS

Newton's Second Law:

$F = ma$

Acceleration:

$a = F \div m$

Weight:

$W = mg$

Torque:

$\tau = F \times d$

Centripetal Force:

$F_c = mv^2 \div r$

Net Force:

$F_{net} = ma$

For balanced forces:

$F_{net} = 0$ N

For balanced torque:

Net torque = 0